# Sutton, Boveri and Chromosomes

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Chromosomes are clearly seen under a standard light microscope (that is one that uses glass lenses) when a cell is undergoing division. (Note: the cell must be stained first). Chromosomes get their names from two Greek words, *chroma* (meaning colour) and *somes* (meaning body), because they absorb certain dyes.

**Cytology:** a branch of biology that deals with the structure and function of cells.

Cytologists had a good idea of the sequences involved in mitosis and meiosis over a hundred years ago.

In the early 1900s biologists began to see that there was a link between the way chromosomes behaved and (the newly rediscovered) Mendel's 'factors'.

Walter Stanborough Sutton was an American physician and geneticist. He (and others) pointed out, in 1902, that Mendel's hereditary factors (genes) occurred in pairs in somatic cells and separated when gametes were formed (gametogenesis). This tied in with the fact that somatic cells contain two of each kind of chromosome and that these chromosomes separate during meiosis. After further study of meiosis (in grasshoppers) he observed that:

- in a cell undergoing meiosis, the chromosomes line up in pairs
- the chromosomes in each pair resemble each other strongly in physical appearance
- at one stage the chromosome pairs separate, giving each resultant gamete (sex cell) only one chromosome from the pair
- when a male sex cell and a female sex cell unite, the resulting zygote has a full complement of chromosomes.

Around the same time, Theodor Heinrich Boveri, a German zoologist, independently demonstrated the separate and continuous existence of chromosomes. Boveri is also noted for his discovery of the centrosome. Between the work of Sutton and Boveri (and other researchers) a chromosome theory of inheritance started to take shape. The points below summarise Sutton and Boveri's views at the time.

- Chromosomes carry the units of inheritance.
- Chromosomes occur in pairs.
- Organisms have many more units of inheritance than there are chromosomes.
- Each chromosome carries hundreds of inheritance factors (genes).

**Centrosome:** the region of clear cytoplasm, near the cell nucleus, that contains centrioles.

Wilhelm Ludvig Johannsen was a Danish botanist and geneticist who came up with the term 'gene' (around 1910) for a unit of heredity. He also was responsible for the distinction of genotype and phenotype.

Further work by Thomas Hunt Morgan (an American embryologist) with *Drosophila melanogaster* (fruit flies) confirmed Mendelism and the chromosome theory, and also led to the conception of the gene as the smallest unit of recombination.

1. What does a *cytologist* study?
2. What is 'gametogenesis'?
3. What is the origin of the name 'chromosome'?
4. Why is the chromosomal theory of inheritance referred to as the 'Sutton–Boveri' theory?
5. What do we now call Mendel's inheritance 'factors'?
6. Outline the views of Sutton and Boveri on the behaviour of chromosomes in meiosis.
7. Which organism was used by T. H. Morgan in his studies?



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